

8.1

$$P_A(x) = \begin{vmatrix} -1-x & -2 & 0 \\ -1 & 3-x & -1 \\ 2 & 3 & 1-x \end{vmatrix} = -x^3 + 3x^2 + 0x - 4 = -(x-2)^2(x+1)$$

$$A + I = \begin{pmatrix} 0 & -2 & 0 \\ -1 & 4 & -1 \\ 2 & 3 & 2 \end{pmatrix}$$

$$\begin{array}{c|ccc|c} & -4 & 0 & 3 & -1 \\ \hline \textcircled{2} & 0 & 2 & 1 & -1 \\ \hline & 2 & 1 & -1 & \\ \hline \textcircled{2} & 0 & -1 & -1 & \\ \hline & -1 & -1 & & \\ \hline \textcircled{-1} & 0 & -1 & & \end{array}$$

$$M_{-1} = LO \left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

$$A - 2I = \begin{pmatrix} -3 & -2 & 0 \\ -1 & 1 & -1 \\ 2 & 3 & -1 \end{pmatrix}$$

$$M_2 = LO \left\{ \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \right\}$$

$$\left(\begin{array}{ccc|c} -3 & -2 & 0 & -2 \\ -1 & 1 & -1 & 3 \\ 2 & 3 & -1 & 5 \end{array} \right) \rightarrow \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \text{ třeba}$$

$$\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \xrightarrow{A-2I} \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \xrightarrow{A-2I} 0$$

 $\bar{u}_1$

$$B_1 = \left(\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} \right)$$

 $\bar{u}_2$

$$B_2 = \left(\begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \right)$$

$$[f_{A/\bar{u}_1}]_{B_1}^{B_1} = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$[f_{A/\bar{u}_2}]_{B_2}^{B_2} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$$

Jelikož máme pouze dva vlastní v., tak z toho poskládáme pouze $\bar{u}_1, \bar{u}_2$. Naskytovala by se i možnost z jor. řetízku. $\bar{u}_3$ s $B_3 = \left(\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} \right)$ to ale nebude fungovat, protože $\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$ není vlastní vektor a není v řetízku s $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$.